

Valentin Py

Aerospace Engineering Student - Computational Fluid Dynamics & Applied Mathematics

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Summary

Aerospace engineering student at ISAE-SUPAERO within the Propulsion and Combustion track, I made an exchange semester focused on numerical methods for fluid dynamics at the University of Tokyo and followed the M2RI at Université Paul Sabatier.

During my previous internships I developed skills about programming (C++/Java/Python), numerical methods for fluid dynamics (Real-time methods SPH/FLIP-PIC or Riemann solvers) and put hands on simulating software for different CFD cases (STARCCM+/OpenFOAM).

My goal is to strengthen my knowledge at the intersection between applied mathematics and numerical methods for CFD. Driven by rigor and a desire to build next-generation simulation tools, I welcome connections with researchers and laboratories in these fields.

Experience

ISAE-SUPAERO - Department of Aeronautics (DAEP), Research & Development Intern - Scientific Visualization & AR/VR

Toulouse, France
Apr 2026 – present
6 months

- Conducted aerodynamic simulations (RANS/URANS) on a NACA0012 airfoil using STAR-CCM+, focusing on flow physics analysis for teaching key concepts of aerodynamics.
- Engineered a real-time VR/AR visualization framework within Unreal Engine 5 using Blueprints & post-processing softwares like Paraview and Blender to render aerodynamic criteria (e.g., Q-criterion, velocity fields, streamlines).
- Developed and explored different numerical methods in C++ for real-time fluid dynamics simulations around NACA0012 airfoil (LBM method through GPU computing).

Ascendance, Software Engineer Intern on ICD

Toulouse, France
Mar 2025 – Aug 2025
6 months

- Solo developed, tested, and integrated a software using Python and JavaFX to manage the Interface Control Documents (ICD) for a hybrid-electric aircraft.
- Designed automated validation and verification (V&V) workflows to bridge the gap between ICD specifications and hardware test benches using custom Python automation scripts.
- Contributed to an innovative thermal-electric propulsion project, strengthening skills in analyzing multidisciplinary data coupling and complex systems engineering.

Fondation ISAE-SUPAERO, Communications & Outreach Officer - OSE ISAE-SUPAERO Program

Toulouse, France
Mar 2024 – Mar 2025
1 year 1 month

Member of the student team leading the OSE (Ouverture Sociale Étudiante) program at ISAE-SUPAERO, responsible for communicating the program's actions and promoting equal opportunities among youth. The OSE program supports more than 1,000 high school and middle school students each year across nearly 30 institutions in the Occitanie region.

- Covered events and interventions aimed at promoting equal opportunities among youth.
- Wrote articles detailing the latest initiatives to ensure visibility of the various projects led by OSE.

Hinfact, Internship - Data Management & Process Digitization

- Structured database and web interfaces to centralize and process training data for airline pilots.
- Architected information management tools (inventories, functional guides, relational databases) using Notion.
- Developed competencies in data curation, information architecture, and process digitization within a fast-paced aerospace tech environment.

Toulouse, France
June 2024 – Aug 2024
3 months

Education

MSc **Université Toulouse III - Paul Sabatier**, Master 2 Recherche et Innovation (M2RI) - Parcours Déterministe, EDP et Simulations Numériques

- Coursework: An Introduction to the Theoretical and Numerical Analysis of Scalar Conservation Laws; Controlled Dynamical Systems: Structured Modelling and Numerical Methods; Regularization of Ill-posed Inverse Problems and Applications; High-Performance Computing (MPI, OpenMP, CUDA).
- Double-degree track with the 3rd-year ISAE-SUPAERO curriculum, focused on rigorous numerical analysis of PDEs applied to CFD.

Toulouse, France
Sept 2026 – Aug 2027

University of Tokyo, Aerospace, Aeronautical and Astronautical Engineering (Exchange Semester)

- Coursework: Advanced Theoretical Numerical Simulation; Discretization Schemes (FDM/FVM); Lattice Boltzmann Method (LBM); Fluid-Structure Interaction (FSI) applied to bioengineering frameworks.
- Academic project: high-fidelity aerodynamic simulations on a NACA 0012 airfoil using OpenFOAM, analyzing mesh convergence and solver stability.

Tokyo, Japan
Sept 2025 – Mar 2026

MEng **ISAE-SUPAERO**, Aerospace, Aeronautical and Astronautical Engineering - Fluid Dynamics / Internal Aerodynamics (Propulsion, Combustion) & Applied Mathematics

- Core coursework: advanced multi-physics fluid dynamics (unsteady, 3D, aeroacoustics, aeroelasticity), numerical simulation and turbulence (instabilities, turbulence modeling, numerical methods).
- Turbomachines & Combustion track: turbomachinery aerodynamics, two-phase flows and combustion, design case studies.
- Technical expertise: hands-on experience with industry-standard CFD software and solvers (Ansys, STAR-CCM+, OpenFOAM) for complex flow analysis and turbulence modeling.

Toulouse, France
2023 – 2027

Projects

SPH Fluid Solver (C++/OpenGL)

Weakly-compressible smoothed-particle hydrodynamics solver (cubic spline kernel, Tait equation of state), validated on the classic dam-break benchmark and rendered in real time with a custom OpenGL engine.

- valentinpy5-spec.github.io/projects/sph_dam_break

Exact Riemann Solver for the 1D Euler Equations

Finite volume solver for the compressible Euler equations with an exact Riemann solver (Newton-Raphson star-region pressure) and RK2 time integration, validated on the Sod shock tube.

- valentinpy5-spec.github.io/projects/1_project

NACA0012 Flow Solver (LBM, GPU)

Real-time Lattice-Boltzmann (D2Q9/BGK) solver in OpenCL for 2D flow around a NACA0012 airfoil, with live angle-of-attack control and OpenGL visualization.

- valentinpy5-spec.github.io/projects/naca0012_lbm_gpu

Extracurricular Activities

OSE ISAE-SUPAERO Program

Toulouse, France | September 2023 - February 2025

- Tutoring and academic mentoring for high-school students (Seconde to Terminale) as part of a "cordées de la réussite" outreach program.

SCUBE Rocketry Club - ISAE-SUPAERO

Toulouse, France | September 2023 - February 2025

- Treasurer of the school's rocketry club.
- Completed a 5-month training program leading to the launch of a mini-rocket with CNES (aerodynamics, propulsion, Python programming).

Arts & Travel Association - ISAE-SUPAERO

Toulouse, France | September 2023 - February 2025

- President of the travel branch, in charge of organizing cultural trips abroad.
- Organized cultural trips in several European countries (Italy, Spain).

Interests

Hobbies

Trail running, Climbing, Swimming

Skills

CFD & Numerical Methods: FDM, FVM, LBM, Galerkin Method, Turbulence Modeling, Mesh Convergence, Scalar Conservation Laws

CFD Software: STAR-CCM+, OpenFOAM, Ansys

HPC & GPU Computing: MPI, OpenMP, CUDA, OpenCL

Scientific Programming: Python (PyVista, NumPy), C++, Java

Scientific Visualization: ParaView, Blender, Unreal Engine 5, OpenGL, AR/VR pipelines

Tools: Git, Cycle en V, Notion

Languages

French: Native speaker

English: Fluent

Japanese: Beginner

German: Beginner